

# DR. MUHAMMAD TAHIR SHEHZAD

## Lecturer in Environmental Sciences • Soil & Environmental Scientist

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### RESEARCH & TEACHING PROFILE

Environmental and Soil Scientist specializing in heavy-metal dynamics (specifically arsenic speciation and mitigation in agro-ecosystems), soil–plant interactions, soil carbon dynamics, mineral–organic interactions, and contaminant remediation. Possesses specialized international research experience, including work at the Global Centre for Environmental Remediation (GCER), University of Newcastle, Australia, utilizing advanced analytical techniques (ICP-MS, IC-ICP-MS). Actively teaches undergraduate agricultural and environmental courses, co-advises M.Sc. and Ph.D. students, and serves on departmental academic and Objective Based Education (OBE) committees.

### SCHOLARLY METRICS

|                                                      |                                        |                                                  |                                                               |                                                 |
|------------------------------------------------------|----------------------------------------|--------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------|
| Total Citations<br><b>&gt; 350</b><br>Google Scholar | h-Index<br><b>10</b><br>Google Scholar | Cumulative IF<br><b>&gt; 60</b><br>Impact Factor | Journal Papers<br><b>23 total</b><br>17 Peer-Reviewed / 6 HEC | Book Chapters<br><b>5</b><br>Published/In Prod. |
|------------------------------------------------------|----------------------------------------|--------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------|

### SELECTED JOURNAL PUBLICATIONS

- Shehzad, M.T.\***, M. Sabir, Saifullah, A.B. Siddique, M.M. Rahman and R. Naidu. 2022. Impact of water regimes on minimizing the accumulation of arsenic in rice (*Oryza sativa* L.). *Water, Air, and Soil Pollution*. 233(9):1-12. (DOI: <https://doi.org/10.1007/s11270-022-05856-7>) [**Impact Factor: 3.0**]
- Shehzad, M.T.\***, M. Sabir, M. Zia-ur-Rehman, M.A. Zia and R. Naidu. 2022. Arsenic concentrations in soil, water and rice grains of rice growing areas of Punjab-Pakistan: Multivariate statistical analysis. *Environmental Monitoring and Assessment*. 194(5):1-16. (DOI: <https://doi.org/10.1007/s10661-022-09852-z>) [**Impact Factor: 3.0**]
- Bashir, M.H., A. Rehman, H.R. Ahmad, A. Hedfi, M.B. Ali, F. Boufahja, K. Elmnasri, E. Mahmoudi and **M.T. Shehzad\***. 2024. Dust trace metals implications on school's indoor air quality linked to human health risk at Khurianwala (Pakistan). *Atmospheric Environment*. 339:120889. (DOI: <https://doi.org/10.1016/j.atmosenv.2024.120889>) [**Impact Factor: 4.2**]
- Rahman, M.M., **M.T. Shehzad**, A.K. Nayak, S. Sharma, M. Yeasmin, S. Samanta, R. Correll and R. Naidu. 2020. Health risks from trace elements in muscles of some commonly available fish in Australia and India. *Environmental Science and Pollution Research*. 27(17):21000-21012. (DOI: <https://doi.org/10.1007/s11356-020-08535-6>) [**Impact Factor: 5.190**]
- Wang, S., L. Wei, L. Zhang, **M.T. Shehzad**, M.A. Hameed, H. Bashir, Q. Ali, M.Z. Hashmi and M.M. Hussain. 2025. Tillage-regulated impacts of engineered Fe/Zn-humic complexes on lead toxicity and soil biochemical health. *International Journal of Phytoremediation*. 28:1-12. (DOI: <https://doi.org/10.1080/15226514.2025.2579150>) [**Impact Factor: 4.003**]
- Bashir, M.H., M. Farhan, T. Samreen and **M.T. Shehzad\***. 2024. Effectiveness of constructed wetland technology-treated industrial wastewater on the spinach (*Spinacia oleracea*) health risks and biochar efficiency. *Environmental Geochemistry and Health*. 46:469. (DOI: <https://doi.org/10.1007/s10653-024-02254-2>) [**Impact Factor: 3.8**]

### SELECTED SCHOLARLY BOOK CHAPTERS

- Shehzad, M.T.\***, H. Ghazanfar, M. Hussain, Z. Farooqi, M.M. Hussain and H. Bashir. 2025. Drought stress influence on tomato: a brief understanding. In: Chaudhry, U.K., Z.N. Öztürk and A.F. Gökçe (eds.). *Drought Stress: Review and Recommendations*. Springer Nature, Switzerland AG. pp. 221-237. (DOI: [https://doi.org/10.1007/978-3-031-80610-0\\_8](https://doi.org/10.1007/978-3-031-80610-0_8))
- Ali, M., M.Z. Ur Rehman, A. Jamil, M.A. Ayub and **M.T. Shehzad**. 2023. Silicon in Soil, Plants, and Environment. In: S. Pandey, D.K. Tripathi, V.P. Singh, S. Sharma and D.K. Chauhan (Eds.). *Beneficial Chemical Elements of Plants: Recent Developments and Future Prospects*. John Wiley & Sons, Inc. pp. 227-255. (DOI: <https://doi.org/10.1002/9781119691419.ch10>)
- Farooqi, Z.U.R., A. Kareem, M.A. Ayub, M.M. Hussain, N. Zeeshan and **M.T. Shehzad**. 2020. Use of pesticides in agriculture: impacts on soil, plant, and human health. In: Mehmood, M.A., K.R. Hakeem, R.A. Bhat and G.H. Dar (Eds.). *Pesticide Contamination in Freshwater and Soil Environs: Impacts, Threats, and Sustainable Remediation* (ed. 1). Apple Academic Press, Palm Bay, Florida USA.

### EDUCATION & PROFESSIONAL SERVICE

#### ACADEMIC CREDENTIALS

- Doctor of Philosophy (Ph.D.) in Environmental Science** (2022)  
*Institute of Soil and Environmental Sciences, University of Agriculture Faisalabad, Pakistan*  
Thesis: Assessment and management of arsenic in rice in Punjab-Pakistan.  
Foreign Research: Global Centre for Environmental Remediation, Univ. of Newcastle, Australia.
- Master of Philosophy (M-Phil) in Environmental Science** (2014)  
*PMAS-Arid Agriculture University, Rawalpindi-Pakistan*  
Thesis: Assessment of heavy metals in soils of urban areas of Rawalpindi.
- Bachelor of Science (Hons.) in Agriculture / Soil Science** (2012)  
*University of Agriculture Faisalabad, Pakistan*

#### ACADEMIC SERVICE & TRAININGS

- Editorial Service:** Review Editor for *Frontiers in Plant-Soil Interactions*.
- Journal Reviewer:** Active reviewer for high-impact journals including *Journal of Hazardous Materials*, *Ecotoxicology and Environmental Safety*, *Environmental Monitoring and Assessment*, *Scientific Reports*, *Pedosphere*, etc.
- International Trainings & Fellowships:**
  - DAAD International Training on GHGs (Geisenheim Univ., Germany, 2024-2025)
  - HEC-5000 Indigenous Ph.D. Scholarship (Phase-II, Batch-IV)
  - IRSIP Fellowship at Univ. of Newcastle, Australia (UoN Honorary Staff)